

SFTP Troubleshooting Guide

Troubleshooting Guide

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Failure-mode playbooks for the enterprise SFTP stack. Each entry: **Symptom** → **Diagnosis** → **Fix** → **Verify**. Verification requires positive output, not absence of errors (§11.4 anti-bluff). Items referencing not-yet-built components cite their ATM item.

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1. Container won't start

Symptom: podman ps shows the sftp container exited; podman-compose up -d reported failure.

Diagnosis:

```
podman logs <sftp-container-name>           # last 50 lines tell the
story
podman-compose -f deploy/docker-compose.yml config # validate
compose syntax
```

Common causes + fixes: - **Malformed users.conf** — every line must match `user:pass[:e][:uid[:gid[:dir...]]]`; a stray space or missing field aborts entryptpoint. Fix the line, re-render via the API (ATM-003 renderer validates before writing), restart. - **Volume**

path missing — create data/ on the host: `mkdir -p data.` - **Port already bound** — see §2. - **Image pull failure** — `podman pull atmoz/sftp:<tag>` separately to see the real error.

Verify: `podman ps --format '{{.Names}} {{.Status}}'` shows Up, and `podman logs` ends with the sshd listening line (no error entries).

2. Port conflicts (7721 / 7722)

Symptom: compose fails with “address already in use”, or connections land on the wrong service.

Diagnosis:

```
ss -tlnp | grep -E ':(7721|7722) '      # who owns the port
```

Fix: stop the conflicting process, or change SFTP_PORT / API_PORT in .env and re-run `service_ctl.sh restart` (ATM-006). Rootless Podman binds as the user — ports <1024 are out of scope by design; stay in the 77xx range.

Verify: `ss -tlnp` shows exactly one listener per port, owned by the expected process; `sftp -P $SFTP_PORT user@127.0.0.1` reaches the SFTP banner.

3. Permission denied on upload

Symptom: login succeeds, `put file.txt` fails with Permission denied.

Diagnosis:

```
ls -ln data/          # numeric UID/GID of the user's home +  
                      subdirs  
podman exec <container> id <user> 2>/dev/null || true
```

Causes + fixes: 1. **Writing into the chroot top-level** — by design impossible; the chroot home is root-owned and non-user-writable (see §4). Upload into the writable subdirectory instead (e.g. `cd upload && put file.txt`). The provisioner (ATM-003) creates it automatically. 2. **UID/GID mismatch** — host directory must be owned by the numeric UID/GID from `users.conf`. Rootless note: in rootless Podman the container UID maps through the user namespace; the project standardizes UID/GID allocation in the renderer so host and container agree. Fix ownership with the matching numeric IDs (as the service user; no sudo in rootless layout — the data root lives under the service user’s home). 3.

Account is read_only — check GET /api/v1/accounts/<user>; read_only rejects uploads by design (ATM-002/003). Change permission if intended (audit-logged).

Verify: put into the writable subdirectory returns 100% progress and ls shows the file with non-zero size; for read_only, the denial itself is the correct verified behavior.

4. Login fails immediately — the chroot write problem

Symptom: client connects, authenticates, then the session closes instantly (Connection closed right after password); server log shows fatal: bad ownership or modes for chroot directory.

Root cause (the classic atmoz/internal-sftp trap): OpenSSH requires the ChrootDirectory (the user's home, %h) to be **owned by root and NOT writable by the user**. If the user's home is user-writable, sshd refuses the session. atmoz/sftp documents the same constraint from the other side: users “can't create new files directly under their own home directory”.

Fix: 1. Ensure the home top-level is root-owned, mode 755, **not** writable by the account's UID. 2. Provide a writable subdirectory inside it (e.g. upload/) owned by the user's UID/GID — this is where clients upload. The ATM-003 directory provisioner does both automatically; manual provisioning must replicate it. 3. Re-render/reload (API reload signal, or service_ctl.sh restart sftp).

Verify: login succeeds; pwd shows /; put into /upload completes; attempted put into / is denied (expected).

5. API authentication failures

PLANNED — STREAM-2 (ATM-002).

Symptom: 401 from /api/v1/**, or login returns 403.

Diagnosis + fixes: - **Wrong/expired token** — tokens are short-lived; re-login via POST /api/v1/auth/login (see [api_quickstart.md](#)). - **Rate-limited** — 429 after repeated failures; back off, check the audit log for the lockout entry; limits are configurable (§6 of security guide). - **Bootstrap already consumed** — a second POST /api/v1/bootstrap returns 409; the super-admin already exists. Use login, not bootstrap. - **Clock skew** — JWT validation fails if host clock drifts; verify timedatectl.

Verify: login returns a token JSON; an authenticated GET /api/v1/accounts returns 200 with a JSON array.

6. Database migration issues

PLANNED — ATM-002 (embedded migrations).

Symptom: API container restarts in a loop; logs show migration failure; /readyz returns non-200.

Diagnosis + fixes: - **Partial migration after crash** — restore the pre-upgrade snapshot (backup.sh restore <id>, ATM-006), then re-run the upgrade; migrations are transactional per step by design. - **SQLite file locked** — only one API instance may own the SQLite file; ensure a single replica and that the volume isn't shared with a stray dev instance. - **Postgres unreachable** — check podman logs <postgres> and the API's DB_* env; readyz distinguishes DB-down from migration-failed.

Verify: /readyz returns 200; the audit log records migration applied: <version> lines.

7. systemd -user / linger issues

PLANNED — ATM-006 units.

Symptom: systemctl --user status sftp-stack errors with “Failed to connect to bus”, or the stack dies at logout.

Diagnosis + fixes: - **No user bus** — you are in a bare SSH session without XDG_RUNTIME_DIR; log in via a proper session (machinectl shell or a real login), or run units with systemctl --user only after loginctl shows your session. - **Stack stops at logout** — user services stop with the last session unless linger is enabled: bash loginctl enable-linger "\$USER" On managed hosts this may require an administrator — it is the only step outside pure user scope. If linger is denied, document the operational constraint and use a supervised foreground run for that host. - **Unit not found** — re-run bash scripts/service_ctl.sh install and systemctl --user daemon-reload.

Verify: systemctl --user is-enabled sftp-stack → enabled; after logout/login, podman ps still shows the containers Up (linger case).

8. SFTP works locally but not from the network

Diagnosis + fixes (from MVP.md, carried forward): -

Firewall: UFW `sudo ufw allow 7721/tcp` — NOTE: this is a host-admin firewall step, distinct from the no-sudo container runtime; on hosts where the operator holds no admin rights, request the port opening from the administrator. `firewalld: sudo firewall-cmd --permanent --add-port=7721/tcp && sudo firewall-cmd --reload.` -

Cloud security group: allow inbound TCP 7721 (or restrict to known client CIDRs). - **Binding:** confirm the port mapping publishes on the right interface (`ss -tlnp`).

Verify: from a second host, `sftp -P 7721 user@<server-ip>` reaches authentication; local test alone is not sufficient proof.

9. MITM / host-key warnings after container recreate

Symptom: clients warn “REMOTE HOST IDENTIFICATION HAS CHANGED” after every container rebuild.

Cause: `atmoz/sftp` generates fresh host keys on first start; ephemeral container storage means new keys every recreate.

Fix: persist `/etc/ssh/host_keys` as a volume (see deployment guide §3 — already in the compose design). Migrate once: recreate with the volume, have clients accept the new key once, done.

Verify: two consecutive container recreates keep the same `ssh-keygen -lf` fingerprint; clients connect without warnings.

10. Related docs

- [deployment_guide.md](#) — install, verify, rollback
 - [user_management_guide.md](#) — permissions, keys
 - [security_guide.md](#) — chroot rule context, hardening
 - [../faq/faq.md](#) — quick answers
 - [docs/research/mvp/MVP.md](#) — baseline troubleshooting (container fails, permission denied, remote connect)
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Sources verified (2026-07-11)

- Internal: docs/research/mvp/MVP.md (permission-denied, container-start, remote-connect entries) · docs/Issues.md ATM-002/003/006 (migrations, provisioner, units).
- External fetched this revision: <https://github.com/atmoz/sftp> (chroot home-not-writable symptom, host-key persistence recommendation).
- OpenSSH ChrootDirectory ownership/mode rule per sshd_config(5) — to re-verify before each release (§11.4.99): man page + atmoz README.